

The unique trace extension is tracial on the whole target

MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm

theorem

Unitary invariance, strong shell sums, and a closed centralizer extend the trace identity beyond the source.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. Choose the state extension $\phi : T \rightarrow \mathbb{C}$ of τ supplied by state extension along ι . Its GNS representation is $(H_\phi, \rho_\phi, \xi_\phi)$, with

$$\|\xi_\phi\| = 1, \quad \phi(t) = \langle \xi_\phi, \rho_\phi(t)\xi_\phi \rangle.$$

The inner product is linear in its second argument. The T -orbit of ξ_ϕ is dense by the GNS construction. The chosen state is tracial: $\phi(ab) = \phi(ba)$ for every $a, b \in T$.

Assumptions

The representative shell family is fixed. The state ϕ is the chosen extension of the CAR trace, not an arbitrary state. Its uniqueness among all extensions has already been established by the separately cited extension-uniqueness theorem.

Conclusion

Every element of T lies in the centralizer of ϕ . Thus the trace identity holds for arbitrary pairs of target elements, not just pairs coming from A .

Strong convergence is used inside the actual GNS representation ρ_ϕ . No norm convergence of shell partial sums in T , and no strong continuity of an arbitrary representation, is asserted.

Proof route

Use extension uniqueness to prove source-unitary invariance. Linear spanning gives the source centralizer; vector-functional limits give the shell generators. The norm-closed star centralizer then contains the generated algebra.

▼ Proof steps

1. For a unitary $u \in A$, put $v = \iota(u)$ and $\zeta = \rho_\phi(v)\xi_\phi$. Unitarity gives $\|\zeta\| = 1$, so $\psi_u(t) = \langle \zeta, \rho_\phi(t)\zeta \rangle$ is a state. Its source restriction is

$$\psi_u(\iota(b)) = \phi(v^*\iota(b)v) = \tau(u^*bu) = \tau(b).$$

The last equality uses the source trace and $uu^* = 1$. Extension uniqueness applies and gives $\psi_u = \phi$, hence $\phi(v^*tv) = \phi(t)$ for all $t \in T$. This is `traceExtension_star_unitary_mul_mul`.

2. Substitute $t = va$ into that invariance equality. Since $v^*v = 1$,

$$\phi(av) = \phi(v^*(va)v) = \phi(va).$$

The four-unitary spanning theorem for a unital complex C^* -algebra writes each $b \in A$ as $b = \sum_{k=1}^4 c_k u_k$. Linearity therefore gives

$$\phi(\iota(b)a) = \sum_k c_k \phi(\iota(u_k)a) = \sum_k c_k \phi(a\iota(u_k)) = \phi(a\iota(b)).$$

The exact application of `CStarAlgebra.exists_sum_four_unitary` is in `traceExtension_shellFamilySourceHom_mul`.

3. For a class j , let $g_j = L_j$ regarded as an element of T , let $w_{j,n}$ be its source shell partial isometries, and set $x_N = \iota(\sum_{n \leq N} w_{j,n})$. The actual-representation shell-sum theorem applies to ρ_ϕ, ξ_ϕ : the vector functional restricts to τ , and the target orbit is dense. It gives $\rho_\phi(x_N) \rightarrow \rho_\phi(g_j)$ strongly.
4. Fix $a \in T$. The source centralizer equality holds for every x_N . Strong convergence at both $\rho_\phi(a)\xi_\phi$ and ξ_ϕ , followed by continuity of $\rho_\phi(a)$ and the inner product, gives

$$\begin{aligned} \phi(x_N a) &= \langle \xi_\phi, \rho_\phi(x_N) \rho_\phi(a) \xi_\phi \rangle \longrightarrow \phi(g_j a), \\ \phi(a x_N) &= \langle \xi_\phi, \rho_\phi(a) \rho_\phi(x_N) \xi_\phi \rangle \longrightarrow \phi(a g_j). \end{aligned}$$

Uniqueness of the scalar limit proves $\phi(g_j a) = \phi(a g_j)$. This is the precise input/output of `vectorFunctional_mul_eq_mul_of_stronglyConverges` in `traceExtension_shellFamilyGenerator_mul`.

5. Define $C_\phi = \{t \in T : \forall a \in T, \phi(ta) = \phi(at)\}$. State positivity supplies $\phi(t^*) = \overline{\phi(t)}$, so C_ϕ is a unital star subalgebra; continuity makes it norm closed. The preceding steps put $\iota(A)$ and all g_j in C_ϕ . The concrete-generation theorem `eq_top_of_source_mem_of_generator_mem` now yields $C_\phi = T$. Substituting any $a \in T$ into its defining property gives $\phi(ab) = \phi(ba)$.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm`
- The fixed concrete target and source map —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- Uniqueness before traciality —
`MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace`

- The target GNS vector functional —
MathlibAnnex.CStarAlgebra.CAR.vectorFunctional_tracialRepresentation
- Conjugation invariance from extension uniqueness —
MathlibAnnex.CStarAlgebra.CAR.traceExtension_star_unitary_mul_mul
- Four-unitary span gives every source element —
MathlibAnnex.CStarAlgebra.CAR.traceExtension_shellFamilySourceHom_mul
- Strong shell sums put generators in the centralizer —
MathlibAnnex.CStarAlgebra.CAR.traceExtension_shellFamilyGenerator_mul
- Shell convergence in the actual cyclic representation —
MathlibAnnex.CStarAlgebra.CAR.stronglyConverges_shell_sums_of_trace_of_cyclic
- The closed star centralizer — MathlibAnnex.Analysis.CStarAlgebra.stateCentralizer
- Closedness of the centralizer —
MathlibAnnex.Analysis.CStarAlgebra.isClosed_stateCentralizer
- Passing the vector-functional identity to a strong limit —
MathlibAnnex.Analysis.CStarAlgebra.vectorFunctional_mul_eq_mul_of_stronglyConverges
- Norm generation of the fixed target —
MathlibAnnex.CStarAlgebra.AtomicConstruction.eq_top_of_source_mem_of_generator_mem

Supporting route explanation

Lean source signature (exact)

```
theorem traceExtension_mul_comm (family : RepresentativeShellFamily)
  (a b : ShellFamilyTarget family) :
  traceExtension family (a * b) = traceExtension family (b * a)
```

In the source	Mathematical meaning
family : RepresentativeShellFamily; ShellFamilyTarget family	The fixed CAR shell family and once-chosen links L_j determine the same concrete target $T = C^*(\pi(A), \{L_j\})$ throughout.
shellFamilySourceHom family; trace	The faithful unital source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, and the normalized CAR trace $\tau : A \rightarrow \mathbb{C}$.
traceExtension family	The chosen state extension $\phi : T \rightarrow \mathbb{C}$ with $\phi \circ \iota = \tau$. It is this extension, rather than an arbitrary state.
a b : ShellFamilyTarget family	Any two target elements $a, b \in T$, not necessarily elements of the source image.
traceExtension family (a * b) = traceExtension family (b * a)	The whole conclusion is $\phi(ab) = \phi(ba)$, reversing the order of these target products. Source-unitary invariance, shell sums and the centralizer belong to the linked proof, not additional input assumptions.

Further source notes: The linked proof's `stateCentralizer` is C_ϕ . Its source and generator membership suppliers carry the unitary argument and strong-limit argument described above; they are separately linked proof results, not extra signature assumptions.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm`

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